



Programming Fundamentals.

C++ I.

LAB5.

Create a lab report for the following programming challenge.

Math Tutor.

Your program must create a menu from which the user can select one of the following operations:

- Modulus (%).
- Sum.
- Product.
- Division. (Avoid division by zero).
- Exit.

The program must create two random numbers and challenge the user to enter the correct answer upon selecting one of the options from the menu. For instance, if the user selects option one, modulus the program should display something like:

What is the modulus of 7 and 9? If the answer is correct the program should display a message like: "Correct" If the answer is wrong the program should display a message like: "Wrong" and then return to the menu from which the user can select another operation.

If the user selects an item that is not listed in the menu, display an error message and display the menu again.

Use proper formatting output tools to enhance presentation to the user.

You must submit two files.

- A pdf document, the lab report: math.pdf.
- A cpp file with your C++ code: math.cpp.

The basic template contains a formula for random numbers:

```
////////////////////////////////////  
y = (rand()%(maxValue-minValue + 1)) + minValue  
////////////////////////////////////
```

A lab report consists of:

- A pdf document where you prove that your program is working as intended. To do this take pictures of your inputs and outputs from your program when it is running. Prove that all options from the menu are working. Use concise comments to document your pictures in the lab report. An example of a lab report is posted under the lessons folder for your reference.

Your lab report must contain the necessary documentation that proves that the following requirements are also met.

Additional Requirements:

- Implement multiplication where one of the factors is one digit number and the other is up to two digits number.
- Implement subtraction so that the user's answer is only positive: ex. Your program should not display 25-46. Use up to two digits for each number.
- Implement modulus operation. Numbers must be up to two digits.

Any questions? Just email me.